BALROG: BENCHMARKING AGENTIC LLM AND VLM REASONING ON GAMES

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ABSTRACT

Existing agentic benchmarks primarily evaluate short-horizon, low-interactivity tasks and are rapidly saturating, masking fundamental deficiencies in long-term planning, exploration, and spatial reasoning that are essential for real-world agency. We introduce BALROG, a unified benchmark aggregating six diverse, long-horizon, procedurally generated game environments: BabyAI, TextWorld, Crafter, Baba Is AI, MiniHack, and NetHack. These environments test navigation, exploration, resource management, credit assignment, dynamics deduction, and long-term planning across tasks ranging from seconds-level human solutions to years-to-master complexity. We evaluate leading language-only and vision-language models in zero-shot settings, revealing significant capability gaps: the best language-only models (Claude 3.5 Sonnet, GPT-4o) achieve only 32% average progress, while vision-language performance often deteriorates further. NetHack remains largely unsolved with even advanced models like o1-preview achieving merely 1.57% progression. Our results expose fundamental deficits in current LLM/VLM decision-making capabilities and provide a rigorous, reproducible testbed to drive progress toward truly agentic systems.

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1 INTRODUCTION

The vision of autonomous AI agents capable of complex, long-term decision-making has captured significant attention, yet current evaluation methods fail to adequately test the core capabilities required for real-world agency. Existing benchmarks like AgentBench (Liu et al., 2023) and WebArena (Zhou et al., 2023) primarily focus on short-horizon tasks with limited interactivity, leading to rapid saturation that masks fundamental deficiencies in the reasoning and planning abilities of large language models (LLMs) and vision-language models (VLMs).

Real-world agency demands robust decision-making over extended horizons—potentially thousands of steps—requiring sophisticated exploration, long-term planning, spatial reasoning, and environmental adaptation capabilities that current evaluations fail to capture.

To address this gap, we introduce BALROG (**B**enchmarking **A**gentic **LLM** and V**LM** **R**easoning **O**n **G**ames), a unified benchmark that aggregates six diverse, long-horizon, procedurally generated game environments. Our benchmark spans a complexity spectrum from simple navigation tasks solvable by humans in seconds (BabyAI) to extremely challenging environments that take years for humans to master (NetHack). This design choice is deliberate: games provide controlled yet rich environments for testing agentic capabilities while offering procedural generation to prevent memorization and ensure reproducible evaluation.

BALROG includes six carefully selected environments: BabyAI for basic navigation and instruction following, TextWorld for text-based exploration and goal achievement, Crafter for resource management and long-term planning, Baba Is AI for rule manipulation and abstract reasoning, MiniHack for tactical dungeon navigation, and NetHack (NLE) (Küttler et al., 2020) for comprehensive strategic gameplay. Each environment tests distinct but overlapping sets of skills essential for autonomous agents (Table 1).

Our comprehensive evaluation of state-of-the-art models reveals sobering results. In language-only settings, the best-performing models (Claude 3.5 Sonnet and GPT-4o) achieve only 32.64% and 32.34% average progress respectively. Vision-language capabilities often deteriorate performance further, with GPT-4o dropping to 22.56%. Most strikingly, NetHack remains largely unsolved,

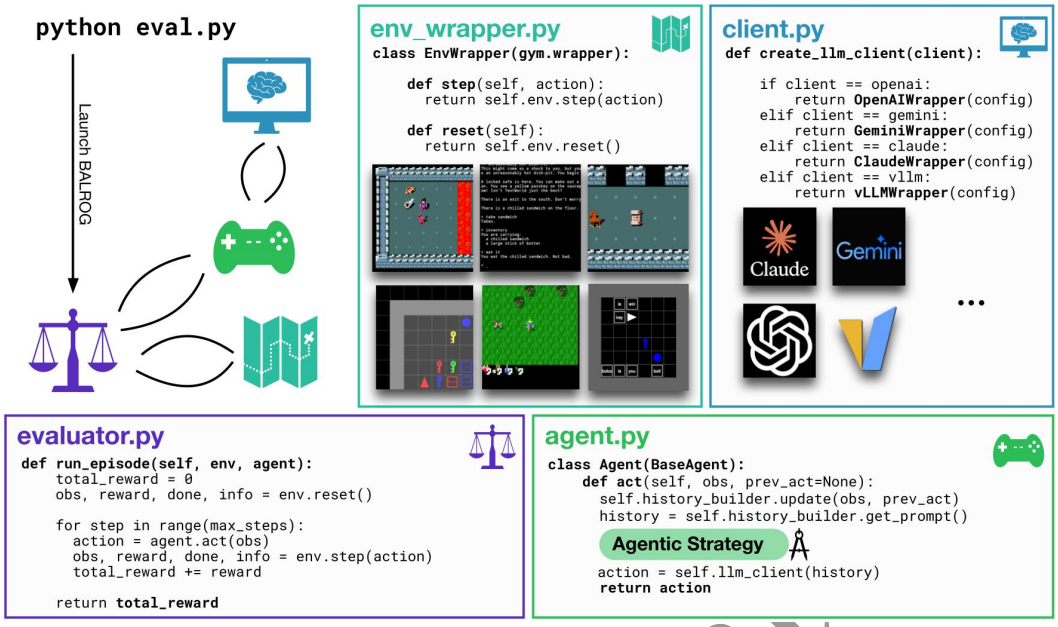

Figure 1: **An overview of the BALROG Benchmark for evaluating LLMs on long-context interactive tasks**. The benchmark provides a unified interface across six diverse game environments, supporting both language-only and vision-language evaluation modes with standardized observation formatting and action handling.

with even the advanced o1-preview model achieving only 1.57% progression, highlighting severe limitations in current long-term planning and decision-making capabilities.

Our contributions are threefold: (1) We present a unified, open benchmark of six long-horizon, procedurally generated environments for comprehensive agentic evaluation; (2) We provide extensive zero-shot baselines across both language-only and vision-language settings for leading closed and open-source models; and (3) We offer an open-source toolkit enabling rapid experimentation with diverse agentic strategies while ensuring standardized, reproducible evaluation.

BALROG reveals fundamental gaps in current AI systems and provides a rigorous path forward for measuring and driving progress toward truly capable autonomous agents.

2 BALROG: Benchmark Design

BALROG unifies six diverse game environments into a comprehensive benchmark for evaluating agentic reasoning capabilities. Our design philosophy centers on three key principles: *long-horizon complexity* to test sustained decision-making, *procedural generation* to prevent memorization, and *skill diversity* to comprehensively evaluate different aspects of agency.

2.1 Environment Selection and Characteristics

We carefully selected six environments that collectively span the spectrum of skills required for autonomous agency, as detailed in Table 1. Each environment contributes unique challenges:

BabyAI (**?**) provides a foundational testbed for basic navigation and instruction following. Tasks involve simple grid-world navigation with natural language instructions, typically solvable in tens of steps. While appearing simple, BabyAI tests fundamental capabilities like spatial reasoning and instruction interpretation that underlie more complex behaviors.

TextWorld (Côté et al., 2019) challenges agents with text-based interactive fiction requiring exploration, object manipulation, and goal completion. The purely textual interface emphasizes language

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understanding and reasoning without visual confounds, typically requiring hundreds of steps to complete.

Crafter (Hafner, 2021) introduces resource management and long-term planning in a Minecraft-inspired environment. Agents must gather resources, craft tools, and survive in a procedurally generated world, demanding thousands of steps and strategic thinking.

Baba Is AI (Cloos et al., 2024) tests abstract reasoning and rule manipulation. Based on the puzzle game "Baba Is You," it requires agents to manipulate rule statements represented as movable objects to achieve goals, testing meta-cognitive abilities.

MiniHack (Samvelyan et al., 2021) offers a curated suite of NetHack-based tasks focusing on specific skills like navigation, combat, and puzzle-solving. These provide stepping stones toward full NetHack gameplay while maintaining procedural generation.

NetHack Learning Environment (NLE) (Küttler et al., 2020) represents the most challenging environment, requiring comprehensive strategic gameplay over tens of thousands of steps. NetHack demands exploration, combat, inventory management, spell casting, and long-term strategic planning, with human experts taking years to achieve mastery.

2.2 Unified Interface and Observation Formats

BALROG provides a standardized interface across all environments, handling the complexity of diverse action spaces and observation formats. Our implementation offers two observation modes:

Language-Only Mode: All environmental observations are converted to natural language descriptions. For originally textual environments (TextWorld, Baba Is AI), we use native text outputs. For visual environments, we provide handcrafted text wrappers that describe the current state, available actions, and relevant environmental features. This mode isolates language reasoning capabilities from visual processing.

Vision-Language Mode: Agents receive both visual observations (rendered images) and corresponding textual descriptions. This mode tests multimodal reasoning and the integration of visual and linguistic information for decision-making.

All environments maintain a consistent history window of 16 turns, providing agents with recent context while managing computational constraints. Initial prompts include environment rules and comprehensive action descriptions to ensure agents understand available operations.

2.3 Evaluation Metrics and Scoring

BALROG employs fine-grained scoring on a 0-100 scale for each episode, moving beyond simple binary success metrics. Scoring varies by environment:

- **Binary Success Environments** (BabyAI, Baba Is AI, MiniHack subtasks): 100 for task completion, 0 otherwise.
- **Proportional Achievement Environments** (TextWorld, Crafter): Scores based on goal completion percentage or achievement milestones.
- **NetHack Progression Metric**: We introduce a novel data-informed progression metric for NetHack, measuring advancement through game milestones rather than simple survival time, providing more meaningful assessment of strategic progress.

Each task is evaluated across multiple random seeds to ensure statistical robustness, with standard errors reported for all results. This design enables fine-grained analysis of model capabilities across different skill dimensions while maintaining reproducible evaluation standards.

3 Zero-Shot Evaluation Protocol

Our evaluation protocol emphasizes zero-shot performance to assess the genuine reasoning capabilities of pretrained models without task-specific optimization. This approach provides insights into

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Table 1: **The tested skills, time horizons, and complexities of interactive decision-making tasks evaluated in BALROG**. Compared to existing benchmarks, BALROG provides infrastructure for evaluating model reasoning and decision-making on harder, longer time-horizon interactive settings. The evaluated tasks span a range of difficulties.

Skills	**BabyAI**	**TextWorld**	**Crafter**	**Baba Is AI**	**MiniHack**	**NLE**
Navigation	✔	✔	✔	✔	✔	✔
Exploration	✔	✔	✔	✔	✔	✔
Resource Management	✗	✔	✔	✗	✔	✔
Complex Credit Assignment	✗	✗	✔	✔	✔	✔
Deducing Env. Dynamics	✗	✗	✗	✔	✔	✔
Long-term Planning	✗	✗	✗	✔	✔	✔
Turns to Complete	10^1	10^2	10^3	10^2	10^2	10^4–10^5
Time to Master for Humans	Seconds	Minutes	Hours	Hours	Hours	Years

the fundamental decision-making abilities that models possess out-of-the-box, which is crucial for understanding their potential as general-purpose autonomous agents.

3.1 Model Selection and Configuration

We evaluate a comprehensive set of state-of-the-art language and vision-language models:

Closed-Source Models: OpenAI GPT-4o and GPT-4o-mini, Anthropic Claude 3.5 Sonnet and Claude 3.5 Haiku, Google Gemini-1.5-Pro and Gemini-1.5-Flash, and OpenAI o1-preview for selected challenging environments.

Open-Source Models: Meta Llama 3.1 (8B, 70B) and Llama 3.2 (1B, 3B, 11B, 90B) variants, providing coverage across different model scales and capabilities.

All models are evaluated using their default configurations with temperature set to 0 for deterministic generation where possible. We use official API endpoints for closed-source models and vLLM (Kwon et al., 2023) for efficient local inference with open-source models.

3.2 Prompting Strategy

Our prompting strategy follows a generic, model-agnostic template designed to provide sufficient context while avoiding task-specific optimizations:

1. **Environment Description:** Clear explanation of the game rules, objectives, and mechanics.
2. **Action Space:** Comprehensive listing of available actions with natural language descriptions.
3. **Observation Format:** Explanation of how environmental state is presented (text-only or multimodal).
4. **History Context:** Previous 16 turns of interaction to provide temporal context.
5. **Current State:** Present environmental observation requiring agent response.

We deliberately avoid chain-of-thought prompting, few-shot examples, or environment-specific optimizations in our baseline evaluation to provide fair comparison across models and establish genuine zero-shot performance.

3.3 Action Handling and Error Recovery

BALROG implements robust action handling to manage the diverse ways models might formulate responses:

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Action Parsing: We employ flexible parsing to extract valid actions from model outputs, accommodating variations in formatting and natural language expression while maintaining evaluation consistency.

Invalid Action Recovery: When models produce invalid actions, the benchmark provides informative feedback messages and executes safe fallback actions (typically no-op or basic movement). This approach allows evaluation to continue while logging errors for analysis.

Timeout Handling: Each episode has environment-specific time limits to prevent infinite loops while allowing sufficient opportunity for task completion.

3.4 Experimental Design

Each model is evaluated across multiple random seeds per environment to ensure statistical reliability. We report mean performance with standard errors across seeds, enabling robust comparison of model capabilities.

Evaluation infrastructure supports highly parallel, asynchronous execution to efficiently process the large number of required episodes. Our implementation maintains strict reproducibility through fixed random seeds and deterministic environment configurations.

We also support optional extensions including few-shot demonstrations and chain-of-thought prompting for future research, though these are not used in our baseline evaluation to maintain zero-shot purity.

4 Results

Our comprehensive evaluation reveals significant limitations in current LLM and VLM agentic capabilities, with performance varying dramatically across environments and modalities. We present results across both language-only and vision-language settings to understand the impact of multimodal processing on decision-making.

Table 2: Language-Only Performance

Model	Average Progress (%)
claude-3.5-sonnet	32.64 ± 1.93
gpt-4o	32.34 ± 1.49
llama-3.1-70b-it	27.88 ± 1.43
llama-3.2-90B-it	27.29 ± 1.44
gemini-1.5-pro	21.00 ± 1.18
claude-3.5-haiku	19.32 ± 1.83
gpt-4o-mini	17.36 ± 1.35
llama-3.2-11B-it	16.82 ± 1.47
llama-3.1-8b-it	15.14 ± 1.55
gemini-1.5-flash	14.63 ± 1.37
llama-3.2-3B-it	10.13 ± 1.28
llama-3.2-1B-it	6.65 ± 1.04

Table 3: Vision-Language Performance

Model	Average Progress (%)
claude-3.5-sonnet	35.48 ± 2.02
gemini-1.5-pro	25.76 ± 1.36
gpt-4o	22.56 ± 1.44
llama-3.2-90B-it	20.99 ± 1.58
gpt-4o-mini	15.36 ± 1.29
gemini-1.5-flash	14.94 ± 1.40
llama-3.2-11B-it	8.43 ± 1.26

4.1 Overall Performance Trends

The aggregate results reveal sobering limitations in current model capabilities for long-horizon agentic tasks. In language-only settings, the best-performing models—Claude 3.5 Sonnet (32.64%) and GPT-4o (32.34%)—achieve less than one-third average progress across environments. This performance ceiling suggests fundamental limitations in current architectures for sustained decision-making.

Model scale shows clear benefits, with larger models consistently outperforming smaller variants. However, even the largest open-source model (Llama 3.2-90B) achieves only 27.29% average progress, indicating that scale alone is insufficient to bridge the capability gap.

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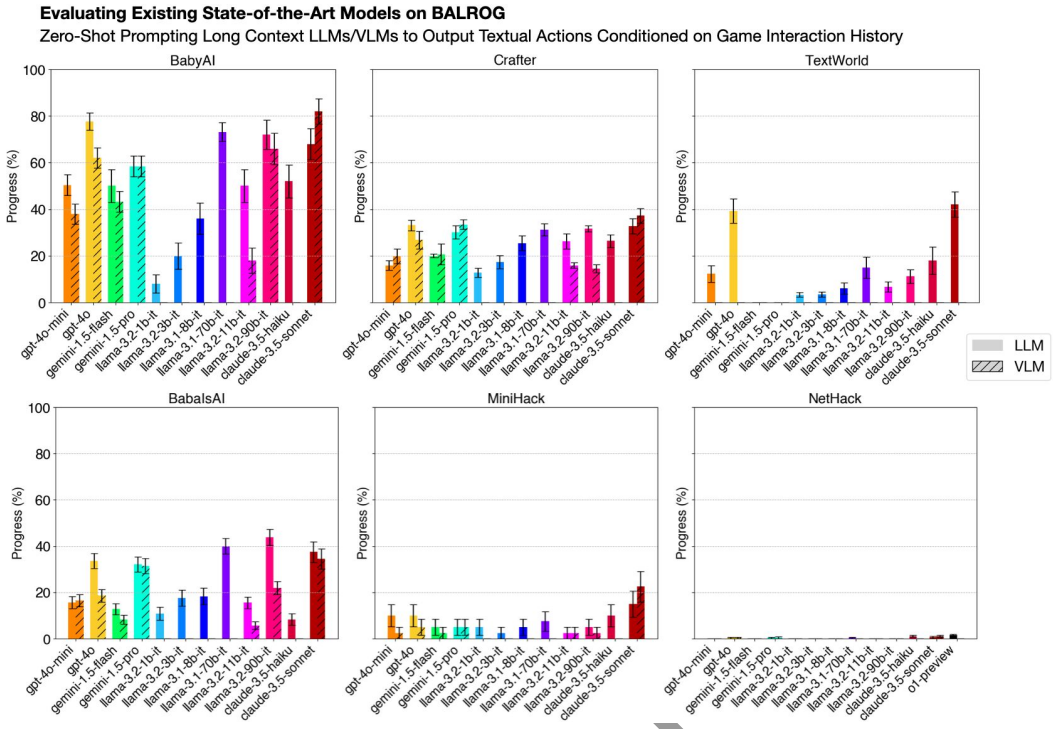

Figure 2: **Baselines for BALROG**. Detailed performance breakdown across all environments for both language-only (LLM) and vision-language (VLM) settings. Results show significant variation in model capabilities across different environments and highlight the challenges posed by complex, long-horizon tasks.

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4.2 Multimodal Performance Paradox

Surprisingly, vision-language capabilities often hurt rather than help performance. GPT-4o drops dramatically from 32.34% to 22.56% when processing visual information alongside text. This counterintuitive result suggests that current VLMs struggle to effectively integrate visual and textual information for sequential decision-making, potentially due to visual processing overhead or interference between modalities.

Notably, Claude 3.5 Sonnet bucks this trend, improving to 35.48% in the vision-language setting, indicating superior multimodal integration capabilities. This performance gap highlights significant differences in how various architectures handle multimodal reasoning.

4.3 Environment-Specific Analysis

Performance varies dramatically across environments, revealing distinct capability profiles:

BabyAI: Most models achieve reasonable performance on basic navigation tasks, with top models reaching 60-80% success rates. This establishes that fundamental instruction-following and spatial reasoning capabilities exist.

TextWorld: Gemini-1.5 models achieve 0% performance due to consistent ”unsafe content” flags triggered by adventure game descriptions, highlighting problematic safety filtering. GPT-4o and Claude 3.5 Sonnet perform best, suggesting superior text-based exploration capabilities.

Crafter: Performance drops significantly, with most models achieving less than 30% progress. The resource management and long-term planning requirements expose severe limitations in strategic thinking.

Baba Is AI: Abstract rule manipulation proves extremely challenging, with models rarely solving puzzles requiring creative rule reinterpretation. This highlights deficits in meta-cognitive reasoning.

MiniHack: The challenging tactical environments remain largely unsolved, with Boxoban and Quest subtasks never completed by any evaluated model, demonstrating failures in complex planning and spatial reasoning.

NetHack: The most striking results emerge from NetHack evaluation. Our novel progression metric reveals that even the advanced o1-preview model achieves only 1.57% progression. Other leading models perform similarly poorly (0-1.5%), with most agents failing to survive beyond early game stages.

4.4 Implications for Agentic AI

These results reveal fundamental gaps between current AI capabilities and the requirements for autonomous agency:

Long-Term Planning Deficits: The dramatic performance drop in complex environments indicates severe limitations in sustained, strategic decision-making over extended horizons.

Exploration vs. Exploitation: Models consistently fail in environments requiring balanced exploration, often getting stuck in local optima or failing to discover necessary resources.

Transfer Learning Limitations: Despite potential exposure to game descriptions during training, models show minimal evidence of effective knowledge transfer to gameplay situations.

Vision-Language Integration: The multimodal performance degradation suggests current architectures struggle with the cognitive load of processing and integrating multiple information streams for sequential decision-making.

These findings underscore the significant research challenges remaining in developing truly capable autonomous agents and validate BALROG as a rigorous benchmark for measuring progress toward genuine agentic AI.

5 Related Work

Our work builds upon and extends several lines of research in agent evaluation, interactive environments, and large language model assessment.

5.1 Agent Benchmarking

Recent efforts to evaluate LLMs as agents have produced several important benchmarks. AgentBench (Liu et al., 2023) provides a multi-dimensional evaluation across eight environments, focusing on reasoning and decision-making abilities. While comprehensive, most AgentBench tasks involve relatively short horizons and limited environmental complexity.

WebArena (Zhou et al., 2023) offers realistic web-based evaluation with functional websites across four domains. Despite its realism, WebArena tasks typically complete in tens of steps and focus primarily on web interaction skills rather than the diverse cognitive abilities required for general agency.

Other specialized benchmarks include ScienceWorld for scientific reasoning tasks, ALFWorld for instruction following in household environments, and IGLU for collaborative building tasks. While valuable, these focus on specific domains rather than providing comprehensive evaluation across diverse agentic capabilities.

BALROG distinguishes itself by focusing specifically on long-horizon evaluation (hundreds to tens of thousands of steps) across diverse skill requirements, providing a more stringent test of sustained decision-making capabilities.

5.2 Game-Based AI Evaluation

Games have long served as testbeds for AI research due to their controlled yet complex nature. Classic examples include Chess, Go, and StarCraft, which have driven advances in search algorithms and reinforcement learning. More recently, procedurally generated environments like NetHack (Küttler

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et al., 2020), Crafter (Hafner, 2021), and BabyAI (**?**) have enabled reproducible evaluation while preventing overfitting to specific scenarios.

TextWorld (Côté et al., 2019) pioneered text-based interactive evaluation, providing procedural generation and customizable difficulty. Baba Is AI (Cloos et al., 2024) introduced meta-cognitive challenges requiring rule manipulation rather than just rule following.

Our contribution synthesizes these diverse environments into a unified evaluation framework, enabling comprehensive assessment across complementary skill dimensions while maintaining the reproducibility and procedural generation that make individual environments valuable.

5.3 Large Language Model Evaluation

Traditional LLM evaluation focuses on static benchmarks like MMLU, HellaSwag, and GSM8K, which test knowledge and reasoning but lack interactive decision-making components. The introduction of GPT-3 (Brown, 2020) and subsequent models has driven interest in more dynamic evaluation approaches.

Recent work has explored LLM capabilities in interactive settings, including code generation, mathematical problem-solving, and conversational AI. However, most evaluations involve single-turn or few-turn interactions rather than the sustained, multi-step decision-making required for autonomous agency.

The emergence of vision-language models has introduced additional complexity in multimodal evaluation. Most existing benchmarks focus on perception tasks (image captioning, visual question answering) rather than vision-guided action selection in interactive environments.

BALROG addresses this gap by providing comprehensive evaluation of both language-only and vision-language decision-making over extended horizons, revealing capabilities and limitations not visible in traditional benchmarks.

5.4 Procedural Content Generation

Our emphasis on procedural generation aligns with recent trends in AI evaluation. Procedural generation prevents memorization of specific solutions while enabling infinite variation for robust statistical evaluation. This approach has proven crucial in reinforcement learning research and is increasingly recognized as essential for fair LLM evaluation.

Each environment in BALROG incorporates procedural generation at different levels: maze layouts in BabyAI, narrative scenarios in TextWorld, world generation in Crafter, puzzle configurations in Baba Is AI, level design in MiniHack, and comprehensive world state in NetHack. This diversity enables evaluation across different types of generalization challenges.

5.5 Multimodal Agent Evaluation

While most agent benchmarks focus purely on language interaction, real-world agency increasingly requires visual understanding. Recent work has begun exploring vision-language agents, but comprehensive evaluation remains limited.

Our dual-modality approach—evaluating the same tasks in both language-only and vision-language settings—provides unique insights into how visual information processing affects decision-making capabilities. The surprising finding that vision often hurts rather than helps performance represents a novel contribution to understanding multimodal agent capabilities.

6 Open Research Problems

BALROG reveals fundamental limitations in current AI systems and highlights several critical research directions for developing truly capable autonomous agents.

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6.1 Long-Horizon Planning and Multimodal Integration

The dramatic performance decline in complex environments exposes severe limitations in long-term strategic planning and multimodal reasoning. Current models struggle to maintain coherent goals while adapting to immediate needs, and paradoxically, visual information often hurts rather than helps performance.

Key challenges include **working memory management** across extended sequences, **hierarchical planning** that decomposes complex goals into sub-goals, and **strategic reasoning** beyond greedy optimization. For multimodal integration, better **attention mechanisms** across modalities, improved **visual representation learning** for decision-making, and **cognitive load management** are needed.

Future research should explore external memory systems, hierarchical planning modules, and specialized training approaches for multimodal sequential decision-making.

6.2 Exploration and Meta-Cognitive Reasoning

Many failures stem from inadequate exploration strategies and limitations in meta-cognitive abilities. Models get trapped in local optima and struggle to reason about rules and systems themselves.

Critical needs include **intrinsic motivation** for curiosity-driven exploration, **systematic exploration strategies** beyond random search, and **transferable exploration knowledge** across domains. For meta-cognition, agents require **rule discovery** mechanisms to infer environmental dynamics, **rule manipulation** capabilities to work around constraints, and **abstraction learning** to identify higher-level patterns.

6.3 Robustness, Evaluation, and Training

The prevalence of invalid actions and safety failures highlights needs for more robust architectures, better evaluation methodologies, and improved training paradigms.

For **robustness**, agents need better **action validity** checking, **failure recovery** mechanisms, and **safe exploration** strategies. **Evaluation improvements** should include more **interpretable metrics** revealing specific capabilities, **human-AI comparisons** across skill dimensions, and **dynamic difficulty adjustment** protocols.

Training paradigms must evolve beyond current approaches through **interactive pretraining** with decision-making data, **multi-step reasoning objectives** rewarding sustained coherence, and better **transfer learning** across domains.

These research directions represent significant challenges but also opportunities for breakthrough advances in autonomous AI systems. BALROG provides a rigorous testbed for measuring progress across these dimensions and driving the field toward more capable agentic AI.

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7 Conclusion

We introduce BALROG, a comprehensive benchmark for evaluating the long-horizon decision-making capabilities of large language models and vision-language models through diverse game environments. By unifying six procedurally generated environments—BabyAI, TextWorld, Crafter, Baba Is AI, MiniHack, and NetHack—BALROG provides rigorous evaluation across the spectrum of skills essential for autonomous agency.

Our extensive evaluation of state-of-the-art models reveals fundamental limitations in current AI systems. Even the best-performing models achieve less than 35% average progress across environments, with particularly poor performance in complex, long-horizon tasks like NetHack (1.57% progression for o1-preview). The counterintuitive finding that vision-language capabilities often hurt rather than help performance highlights critical gaps in multimodal reasoning for sequential decision-making.

These results underscore that current AI systems, despite impressive performance on traditional benchmarks, remain far from the capabilities required for real-world autonomous agency. The chal-

lenges exposed by BALROG—long-term planning, robust exploration, multimodal integration, and strategic reasoning—represent critical research directions for the field.

BALROG provides the research community with a unified, reproducible testbed for measuring progress toward truly agentic AI systems. Our open-source implementation enables rapid experimentation with diverse approaches while ensuring standardized evaluation. As the field continues to pursue autonomous AI capabilities, benchmarks like BALROG will be essential for identifying genuine advances and guiding research toward the most impactful directions.

The journey toward capable AI agents is clearly still in its early stages. BALROG offers both a sobering assessment of current limitations and a clear path forward for measuring and achieving progress in this crucial area of AI research.

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